

Probabilistic Optimization and Forecasting of Nuclear Power Plant Construction

A Model Based on Probabilistic Planning,
Multi-Criteria Optimization, and Machine Learning

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Summary based on the original research paper:

Mathematical Optimization and Forecasting of Nuclear Power Plant Construction Progress using Probabilistic Models and Machine Learning Methods

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1 Purpose

The model represents a nuclear power plant construction project not as a linear schedule with fixed dates, but as a space of possible project execution trajectories. Each trajectory is described by a set of parameters, including schedule, cost, risk, quality, and resources.

The parameters of each activity are defined not by a single value, but by a range of possible values with associated probabilities. This makes it possible to move from a single deterministic planning value to probabilistic estimates, such as the probability of completing the project by a specified date, within a specified budget, and with an acceptable level of risk.

The model represents the trade-off space between these parameters and enables simultaneous optimization of the project trajectory according to multiple criteria.

The model integrates network planning methods (PERT, GERT, PDM), statistical simulation (Monte Carlo), formal decision-making criteria, and machine learning algorithms into a unified system.

2 Main Tasks

The model addresses four interconnected tasks:

- **Identification of critical paths under uncertainty.**

Repeated simulation is used to identify all possible critical and subcritical paths together with their probabilities of occurrence. Instead of a single fixed critical path, the model produces a

set of possible paths with probability estimates.

- **Decision-making at branching points.**

At nodes where alternative project continuation scenarios are possible, formal decision criteria are applied. Each alternative is evaluated using a combination of parameters such as schedule, cost, risk, and quality. The selected alternative depends on the acceptable level of risk and the project objectives.

- **Multi-criteria optimization.**

Optimization is performed simultaneously according to several criteria: schedule, cost, risk, and quality. The model constructs a frontier of feasible solutions, allowing management to evaluate trade-offs, for example, accelerating the schedule through additional expenditure or reducing risk by increasing reserves.

- **Machine learning for forecasting.**

Algorithms are trained on accumulated project data to forecast activity parameters such as duration and cost, classify risks, and detect anomalies.

3 Multidimensional Project Space

The project is represented as a directed graph in which each state is characterized not only by time, but also by a set of attributes such as accumulated cost, resources used, risk level, and degree of readiness. Connections between states represent permissible transitions with associated probabilities and costs.

Each activity is characterized by three duration estimates: optimistic, most likely, and pessimistic. Similar estimates are used for cost and risk.

At the initial stage, standard PERT estimates are used. As actual project data accumulate, starting from approximately 30 observations, the model transitions to probability distributions derived from actual project statistics.

To represent processes with alternative outcomes, the model uses GERT, an extension of network modeling in which transitions between nodes are assigned probabilities. At each branching node, the sum of the probabilities of outgoing transitions equals one.

This enables modeling of situations in which the outcome of an activity is not known in advance. For example, a test may be completed successfully or may require repetition; an acceptance procedure may succeed on the first attempt or may result in comments requiring corrective action.

GERT subgraphs are integrated into the overall model structure and are used to analyze processes with uncertain outcomes.

Monte Carlo simulation repeatedly generates random values for the parameters of all activities and determines the critical path, total cost, and risk profile for each simulated case.

As a result, probability distributions are obtained for key project parameters, including completion dates, total cost, and aggregate risk.

This makes it possible to model alternative project execution trajectories, evaluate parameters along each trajectory, identify alternative routes for completing the work, and perform simultaneous optimization according to schedule, cost, risk, and quality.

4 Decision-Making Criteria

At branching points where alternative scenarios are possible, the model applies formal decision-making criteria. Each scenario is evaluated using a set of parameters such as schedule, cost, risk, and quality. The choice of criterion depends on the decision context:

- **Wald criterion** — focuses on the worst-case scenario. It is applicable to decisions involving safety and regulatory requirements.
- **Savage criterion** — minimizes the possible regret associated with a decision. It is applicable when selecting among strategies under limited information.
- **Bayes criterion** — optimizes the expected average outcome. It is applicable when reliable statistical information on scenario probabilities is available.
- **Hurwicz criterion** — balances optimism and pessimism using an adjustable parameter. It is applicable under partial uncertainty and where flexible planning is required.

These criteria can be applied at three levels: centrally for strategic planning, locally at branching nodes, and adaptively using forecasts generated by machine learning algorithms.

5 Multi-Criteria Optimization

5.1 Local Optimization of Project Sections

Within a selected section of the project network, the model optimizes the allocation of resources among activities, taking into account their contribution to schedule, cost, and risk.

For each activity within the selected section, an optimal combination of parameters is determined using mathematical optimization methods.

5.2 Iterative Optimization of Critical Paths

Optimization is performed iteratively.

At each step, the current critical path is optimized according to a combination of parameters including schedule, cost, risk, and quality.

All paths are then recalculated. If a subcritical path becomes more significant, the optimization process switches to that path.

The process continues until a conditionally optimal project plan is obtained.

5.3 Optimization in Multidimensional Space and the Pareto Front

Each project trajectory is evaluated simultaneously according to four indicators:

- completion time;
- cost;
- risk;
- quality.

The optimization problem consists of improving these four indicators simultaneously.

Several methods can be used:

- **Weighted aggregation.**

Multiple criteria are combined into a single objective function using weighting coefficients. This makes it possible to define priorities, for example, specifying that schedule performance is twice as important as cost. Each set of weights produces a corresponding optimal trajectory.

- **Pareto optimization.**

A set of solutions is constructed in which no trajectory is better than another across all criteria simultaneously. Each solution represents a specific trade-off, for example, completing the project three months earlier at an 8% higher cost, or reducing risk by 15% while extending the schedule by two months.

Management can select a solution depending on project priorities.

- **Multi-criteria graph search.**

Optimal paths are identified using all relevant parameters rather than schedule alone.

- **Machine-learning-based optimization.**

Algorithms identify relationships between project parameters and use them to accelerate the search for optimal solutions.

This enables management to evaluate practical trade-offs and make informed decisions, such as determining the cost of accelerating the schedule by one month, estimating the risk associated with cost reductions in a specific project area, or assessing how changes in quality requirements affect schedule and budget.

6 Machine Learning

6.1 Data Space

Within the project graph, each state is described by time and a set of attributes such as cost, resources, risk, and readiness.

Connections between states represent permissible transitions with associated probabilities and costs.

Machine learning algorithms are trained and applied using these data.

6.2 Formation of Training Data

Training data are obtained from two sources.

The first source is the output of Monte Carlo simulations, which makes it possible to begin model training before a sufficient volume of actual project statistics has been accumulated.

The second source is actual project data as they become available.

As additional actual data are accumulated, the models are retrained and the forecasts are refined.

6.3 Application Areas

Machine learning algorithms are used in four principal areas:

- **Forecasting activity parameters** — duration, cost, and resource requirements based on the characteristics of similar completed activities.
- **Risk classification** — based on activity type, current project status, deviation history, and resource utilization.
- **Anomaly detection** — identification of deviations from typical parameter values that may indicate emerging problems before they become critical.
- **Optimization recommendations** — generation of recommendations for selecting scenarios at branching points based on accumulated project experience.

6.4 Clustering

Clustering methods group activities according to similarity of their characteristics, identify recurring execution patterns, and detect activities that fall outside typical groups.

This helps identify systemic problems that may not be apparent when individual activities are analyzed separately.

6.5 Adaptability

As the project progresses, the algorithms are retrained using newly accumulated data, increasing forecast accuracy.

The model monitors the quality of its own forecasts and, when accuracy decreases, indicates the need for model adjustment.

7 Conclusion

The model integrates probabilistic planning, formal decision-making criteria, multi-criteria optimization, and machine learning into a unified adaptive management system for nuclear power plant construction projects.

It provides an integrated framework covering the principal project parameters, including schedule, cost, risk, and quality.

The model recalculates forecasts as actual data become available, optimizes project trajectories according to multiple criteria, and supports decision-making under uncertainty.

It also generates datasets that can be used for subsequent model training and refinement.

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